CLINICAL · NOSE & SINUSES

External nose and nasal vestibule

The external nose sits in the middle of the face, so its problems are seen by everyone: infections of the skin and vestibule, deformities after injury, the bulbous rhinophyma of rosacea, and skin cancers that love the sun-exposed tip and ala. Two rules stand out: never squeeze a nasal furuncle, and biopsy any lesion that does not heal.

AT A GLANCE

- A furuncle can spread to the cavernous sinus: antibiotics, never squeeze.
- Saddle, hump and crooked nose have different causes.
- Rhinophyma is acne rosacea of the nose.
- Basal cell carcinoma is the commonest cancer of the nasal skin.

1 CHAPTER ONE Overview

Skin, cartilage and the vestibule.

1.1 Applied anatomy

The upper third of the nose is bony (paired nasal bones and the frontal processes of the maxillae); the lower two-thirds is cartilaginous (upper lateral cartilages, lower lateral or alar cartilages, and the septal cartilage). The skin is thin and mobile over the dorsum and thick and sebaceous over the tip and alae. The *vestibule* is the skin-lined entrance, with hairs (vibrissae), up to the limen nasi.

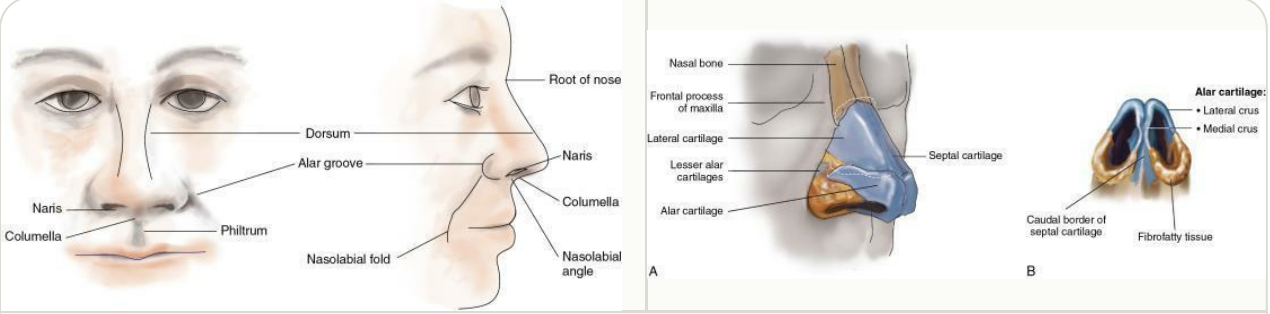


FIGURE 1 First, the parts of the nose and related facial structures. Second, the osteocartilaginous framework.

Dhingra, figs 23.1 and 23.2

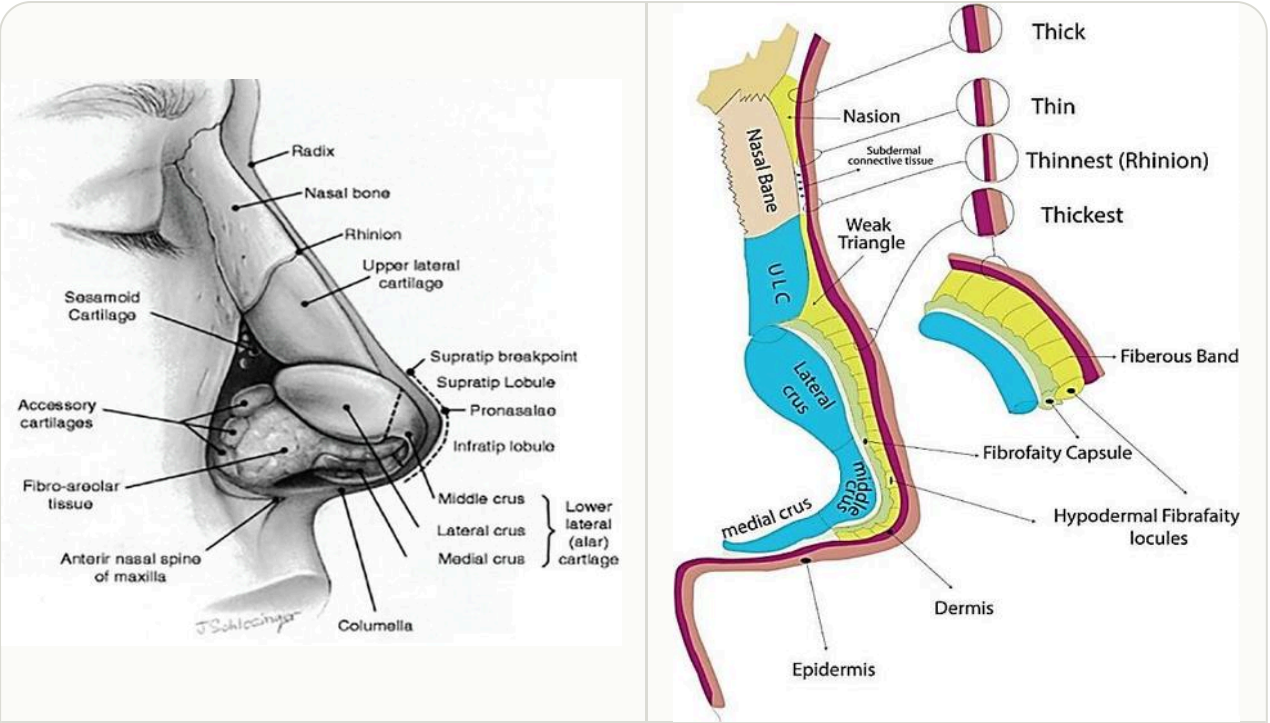


FIGURE 2 The osteocartilaginous framework of the nose, and the change in skin thickness from nasion to tip.

Anatomy atlas · ENT Steps p. 93, ENT Steps p. 96

1.2 The danger area

Veins of the upper lip and nose drain through the facial and angular veins and the ophthalmic veins to the cavernous sinus, without valves. Infection here can spread to the cavernous sinus.

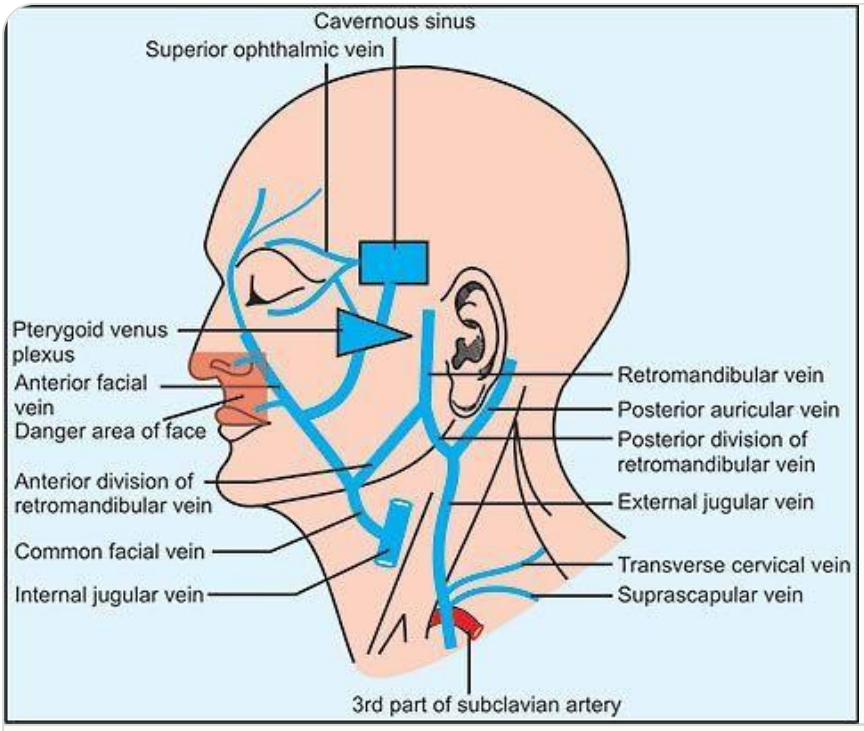


FIGURE 3 The danger area of the face and its venous drainage to the cavernous sinus.
Anatomy atlas · ENT Steps p. 96

2 CHAPTER TWO Infections

Skin and vestibule.

2.1 Furuncle of the vestibule

2.1.1 Cause

Acute staphylococcal infection of a hair follicle, usually after nose picking or plucking hairs.

2.1.2 Clinical picture

A small, exquisitely tender red swelling in the vestibule, with redness and swelling spreading to the tip and dorsum; it may point and rupture.

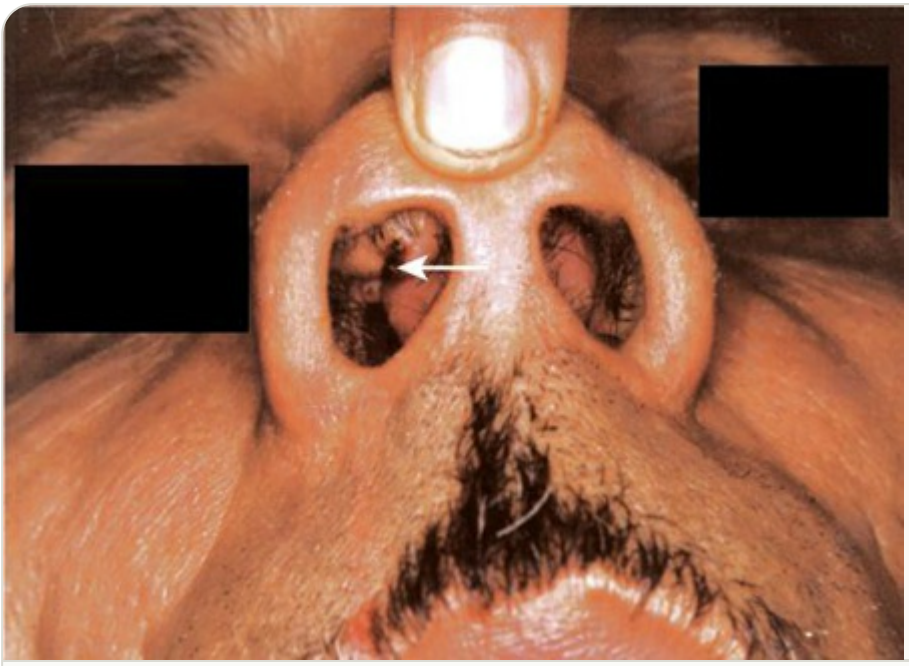


FIGURE 4 Furuncle of the right nasal vestibule.
Dhingra, fig. 25.9

2.1.3 Treatment

Warm compresses, analgesia, and topical and systemic antistaphylococcal antibiotics; incise only when fluctuant.

NEVER SQUEEZE A NASAL FURUNCLE

Squeezing or early incision can spread infection through the angular and ophthalmic veins to the cavernous sinus. Other complications: cellulitis of the upper lip and septal abscess. Recurrent furuncles mean checking for diabetes and nasal staphylococcal carriage.

2.2 Vestibulitis

2.2.1 Cause

Diffuse dermatitis of the vestibular skin, usually *Staphylococcus aureus*, encouraged by nasal discharge from rhinitis or sinusitis and the trauma of wiping.

2.2.2 Clinical picture

Acute: red, swollen, tender skin with crusts over erosions, sometimes involving the upper lip. Chronic: thickened skin with painful fissures and crusts.



FIGURE 5 Acute vestibulitis on the left side.
Dhingra, fig. 25.10

2.2.3 Treatment

Clean the crusts with hydrogen peroxide, apply an antibiotic-steroid ointment and continue it for several days after apparent cure; cauterise a chronic fissure with silver nitrate; treat the source of discharge.

2.3 Cellulitis

Streptococcal or staphylococcal infection making the nose red, swollen and tender, sometimes spreading from the vestibule; systemic antibiotics, warmth and analgesia.

2.4 Specific infections

Lupus vulgaris (apple-jelly nodules), leprosy and syphilis also affect the external nose (see the granulomatous disease topic).

3 CHAPTER THREE Deformities

Shape and function.

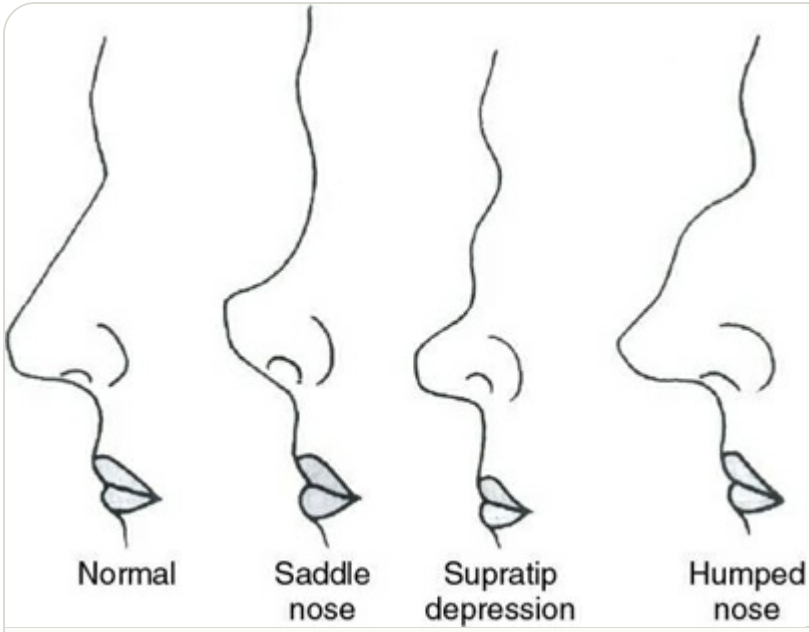


FIGURE 6 Deformities of the nose.
Dhingra, fig. 25.1

3.1 Saddle nose

A depressed dorsum, bony, cartilaginous or both.

3.1.1 Causes

Depressed fractures (the commonest), excessive removal of septal cartilage at submucous resection, septal haematoma or abscess, granulomatosis with polyangiitis, cocaine, leprosy, tuberculosis, syphilis and relapsing polychondritis.

3.1.2 Treatment

Augmentation rhinoplasty: septal, auricular or rib cartilage for a cartilaginous depression; bone (iliac crest or calvarial) for bony and cartilaginous loss. Autografts are preferred; synthetic implants tend to extrude.

3.2 Hump nose

A dorsal hump of bone, cartilage or both, corrected by *reduction rhinoplasty*: removal of the hump and lateral osteotomies to close the open roof.

3.3 Crooked and deviated nose

In a *crooked* nose, the dorsum from the root to the tip is C- or S-shaped; in a *deviated* nose, the dorsum is straight but lies to one side. Both usually follow injury, often in childhood. Correction by *septorhinoplasty* restores both appearance and airway.

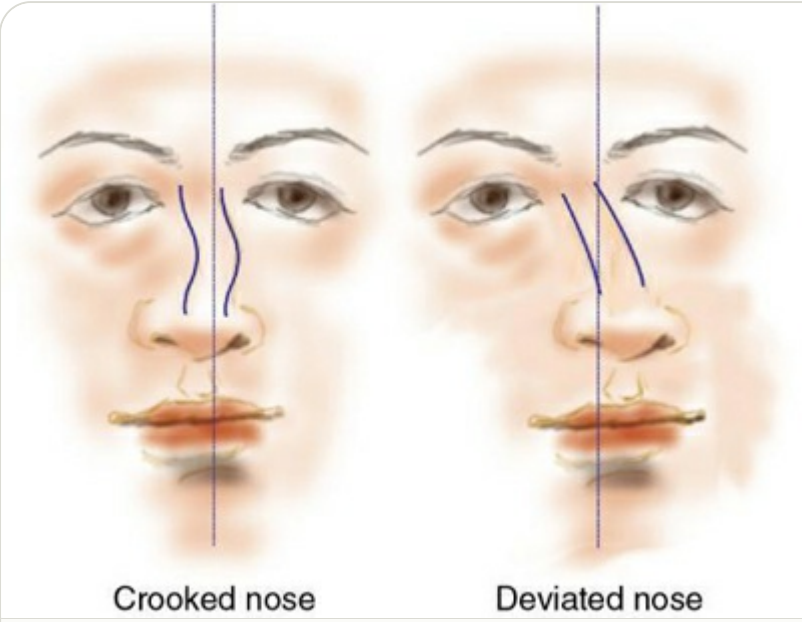


FIGURE 7 The nasal bridge is S-shaped in a crooked nose, and straight but displaced in a deviated nose.
Dhingra, fig. 25.2

3.4 Nasal valve collapse

Weak or scarred lateral walls collapse on inspiration; Cottle's test is positive. Spreader grafts, alar batten grafts or functional rhinoplasty.

OPERATIVE GUIDES IN THEATRE 3 operations

Rhinoplasty and Surgery of the Nasal Valve
Rhinoplasty and the nasal valve

Rhinectomy
Rhinectomy for cancer

Nasal Reconstruction
Nasal reconstruction

CHAPTER FOUR

4 Tumours of the external nose

Benign and malignant.

4.1 Benign

Papilloma (wart), haemangioma, pigmented naevus, seborrhoeic keratosis, neurofibroma and sweat gland tumours. Congenital masses (dermoid, glioma, encephalocele) are covered in their own topic.

4.2 Rhinophyma

The *potato nose*: a slow, lobulated, pink overgrowth of sebaceous glands and connective tissue over the tip and alae, with dilated vessels, in men past middle age with long-standing acne rosacea. It is cosmetic, but may block breathing or vision. Treatment is shaving it down with a scalpel, dermabrasion or carbon dioxide laser, letting the surface re-epithelialise from the remaining glands.



FIGURE 8 Rhinophyma.
Dhingra, fig. 25.4

4.3 Basal cell carcinoma

The commonest malignant tumour of the nasal skin (about 87%), at 40 to 60 years, on the tip and ala: a pearly nodule or an ulcer with rolled edges (*rodent ulcer*). Slow, locally destructive, almost never metastasises. Excision with 3 to 5 mm margins, Mohs surgery for high-risk sites and recurrences, or radiotherapy or cryotherapy for early lesions; reconstruction with local flaps.



FIGURE 9 Basal cell carcinoma of the nose.
Dhingra, fig. 25.6

4.4 Squamous cell carcinoma

The second commonest (about 11%), an infiltrating nodule or ulcer with everted edges on the side of the nose or columella; nodal metastases in about 20%. Radiotherapy for early lesions; wide excision with reconstruction, and neck dissection for nodes.



FIGURE 10 Squamous cell carcinoma of the nose, and of the nasal tip.
Dhingra, fig. 25.7

4.5 Melanoma

The least common: superficial spreading or nodular; wide excision.

4.6 Total rhinectomy

For advanced cancers, followed by a prosthesis or a forehead flap reconstruction.

5 CHAPTER FIVE The vestibule

Narrowing and swellings.

5.1 Stenosis and atresia of the nares

Trauma, burns, surgery, destructive inflammation (formerly smallpox) and Young's operation can narrow the nostrils; congenital atresia from failure of canalisation is rare. Corrected by excision of scar with flaps or grafts and stenting.



FIGURE 11 Stenosis of the left naris after smallpox (A) and after trauma (B).
Dhingra, fig. 25.11

5.2 Nasoalveolar (nasolabial) cyst

A smooth swelling in the floor and lateral wall of the vestibule, raising the ala and filling the nasolabial fold; excised through a sublabial incision.



FIGURE 12 A nasoalveolar cyst at operation (arrow).
Dhingra, fig. 25.12

5.3 Vestibular papilloma and carcinoma

Warts are excised. Squamous carcinoma of the vestibule may spread to the floor,

columella and upper lip, and to the parotid and submandibular nodes.

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Furuncle:** the danger area, and why not to squeeze.
- **Vestibulitis:** cause and treatment.
- **Causes of saddle nose,** and its correction.
- **Crooked versus deviated nose.**
- **Rhinophyma.**
- **Basal versus squamous cell carcinoma** of the nose.

6.2 Common pitfalls

COMMON PITFALLS

- Squeezing or incising an early furuncle.
- Treating a non-healing ulcer with creams without biopsy.
- Using synthetic implants that later extrude.

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IMAGING

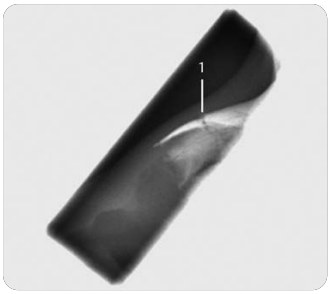
On imaging

The imaging of this topic: key findings, differentials and representative scans.

NOSE & NASAL CAVITY · TRAUMA

Fracture of the Nasal Bone and Retention Cysts

- In cases of large cysts resulting in obstruction of the maxillary sinus and those accompanied by signs of sinusitis, surgical removal (marsupialization) might be considered to improve drainage.



Plain radiograph, lateral view. This view of the nasal bone in a patient with facial trauma demonstrates a fracture of the nasal bone (1), with slight…